

BOLOGOYE, Russian Federation

WMO: 262980

Lat: 57.883N

Lon: 34.033E

Elev: 612

StdP: 14.37

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.6	-6.9	-17.9	2.1	-13.5	-11.2	3.1	-6.0	17.1	27.6	14.4	27.1	2.1	320	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.5	83.1	67.5	79.5	65.9	76.2	64.1	69.4	79.9	67.4	76.4	65.7	73.6	4.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.7	97.0	73.7	64.1	91.7	71.3	62.7	87.0	70.0	33.8	80.3	32.2	76.8	30.9	73.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.2	12.1	10.8	-20.7	86.8	6.0	4.0	-25.0	89.7	-28.5	92.0	-31.9	94.3	-36.3	97.2		
			-20.9	71.9	6.0	2.7	-25.2	73.8	-28.7	75.4	-32.1	76.9	-36.4	78.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.9	18.3	18.7	28.0	41.3	52.6	60.0	65.2	60.8	51.2	40.5	30.1	23.1	(6)
	DBStd	18.63	12.54	12.51	8.52	8.65	8.63	6.85	6.03	6.34	6.71	7.48	9.21	11.60	(7)
	HDD50	4707	981	876	682	287	72	4	0	2	64	307	597	835	(8)
	HDD65	8930	1446	1296	1147	711	394	183	70	161	414	761	1047	1300	(9)
	CDD50	1403	0	0	0	26	151	303	472	338	102	11	0	0	(10)
	CDD65	150	0	0	0	0	9	32	76	32	1	0	0	0	(11)
	CDH74	1232	0	0	0	2	84	276	592	265	13	0	0	0	(12)
	CDH80	284	0	0	0	0	14	58	150	62	0	0	0	0	(13)
(14) Wind	WSAvg	4.7	5.3	5.1	5.3	4.7	4.6	4.0	3.5	3.8	4.2	4.8	5.4	5.6	(14)
(15) Precipitation	PrecAvg	26.8	1.9	1.4	1.4	1.4	2.3	3.1	3.4	3.1	2.3	2.5	2.1	1.9	(15)
	PrecMax	34.5	3.5	2.2	2.3	2.6	4.1	5.8	10.9	6.2	5.2	5.0	4.2	3.1	(16)
	PrecMin	16.7	0.6	0.4	0.3	0.5	0.9	0.8	0.7	0.2	0.7	0.6	0.1	0.7	(17)
	PrecStd	4.1	0.7	0.5	0.5	0.6	0.9	1.3	2.3	1.5	0.9	1.1	0.9	0.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.3	40.1	53.9	73.0	81.8	85.8	89.7	86.4	76.1	63.3	48.5	44.8	(19)
		MCWB	38.1	37.0	43.1	56.3	65.2	67.2	71.1	70.1	63.9	55.7	46.3	43.8	(20)
	2%	DB	35.8	37.3	46.6	66.4	75.7	81.1	84.0	80.2	69.1	56.3	45.9	40.9	(21)
		MCWB	34.4	34.8	38.5	51.9	61.2	65.6	68.4	66.0	59.7	52.0	44.2	40.0	(22)
	5%	DB	34.1	35.1	42.3	61.1	71.3	76.9	80.1	76.2	64.8	53.5	43.4	37.9	(23)
		MCWB	32.9	33.3	36.8	49.3	58.3	63.8	67.1	64.5	57.5	50.7	41.9	36.7	(24)
	10%	DB	32.9	33.7	38.9	56.5	67.2	73.0	76.7	72.4	61.5	51.0	41.0	35.6	(25)
		MCWB	31.9	32.3	35.2	46.8	56.2	62.2	65.2	63.0	56.1	48.8	39.7	34.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.4	38.0	44.1	57.2	67.4	70.1	72.5	71.7	64.6	56.0	47.3	43.6	(27)
		MCDB	39.3	39.4	53.2	71.9	78.4	80.1	84.5	83.7	73.4	60.5	48.2	44.5	(28)
	2%	WB	34.9	35.1	40.1	53.2	63.7	67.3	70.0	68.4	61.2	53.0	44.5	40.0	(29)
		MCDB	35.4	36.4	44.3	62.9	72.9	77.4	81.3	77.0	66.7	55.6	45.5	40.8	(30)
	5%	WB	33.2	33.7	37.6	50.5	60.5	65.6	68.1	66.0	58.9	51.1	42.1	36.9	(31)
		MCDB	33.9	34.9	41.1	59.4	68.3	74.6	77.7	73.2	63.4	53.3	43.2	37.8	(32)
	10%	WB	31.9	32.4	35.6	47.8	57.6	63.3	66.4	63.9	56.6	48.8	39.8	34.6	(33)
		MCDB	32.8	33.6	38.5	55.7	65.1	70.7	74.2	70.7	60.9	50.9	40.8	35.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	11.5	13.5	16.4	17.6	16.4	16.5	15.9	14.0	9.6	6.8	7.3	(35)
		MCDBR	6.1	7.7	18.4	24.0	23.3	21.8	21.5	22.0	19.4	13.3	8.4	6.0	(36)
	5% WB	MCWBR	6.0	6.7	12.2	12.5	10.7	9.5	8.5	9.7	10.7	9.0	7.7	6.1	(37)
		MCWBR	6.2	6.7	14.8	22.1	20.6	18.3	19.5	18.8	16.7	11.4	7.7	5.9	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.276	0.313	0.332	0.379	0.372	0.363	0.393	0.392	0.356	0.336	0.308	0.265	(40)
		taud	2.123	2.106	2.169	2.256	2.386	2.435	2.362	2.353	2.433	2.403	2.235	2.111	(41)
	Ebn at Noon	Ebn at Noon	176	222	255	262	273	278	265	256	249	219	164	141	(42)
		Edh at Noon	17	27	33	36	34	33	35	33	26	21	15	12	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	130	370	833	1251	1625	1675	1636	1296	806	346	117	70	(44)
	RadStd	RadStd	24	67	103	143	142	159	210	117	107	71	19	17	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (9 neighbors)	+1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page